

Mutual Fund Analytics Platform

Data Ingestion & ETL · Exploratory Data Analysis · Fund Performance & Risk Analytics
Advanced Risk Modeling (VaR/CVaR) · Power BI Dashboard

Project	Capstone Project I — Mutual Fund Analytics
Organization	Bluestock Fintech Internship
Cohort	MJ28
Submitted By	balugurisruthi61
Submission Type	Individual Submission
Repository	github.com/balugurisruthi61-bot/data-project
Submission Date	June 14, 2026
Status	COMPLETED

Executive Summary

This report documents the complete data analytics workflow built for the Bluestock Fintech Capstone Project I — Mutual Fund Analytics. As an individual contributor, I designed and implemented a Python-based data pipeline, performed exploratory data analysis across ten structured datasets, computed fund performance and risk metrics, and built a Power BI dashboard for investor transaction analytics.

KEY OUTCOME: Built an end-to-end pipeline covering 40 mutual fund schemes across 10 fund houses, 46,000 NAV records spanning 4.4 years, and 32,778 investor transactions — with computed CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, VaR, CVaR and a live Power BI dashboard.

Project Scope & Objectives

- ✓ Fetch live NAV data for 5 large-cap schemes via the AMFI mfapi.in API (live_nav_fetch.py)
- ✓ Ingest and cross-validate fund master data against NAV history for data quality (data_ingestion.py)
- ✓ Perform EDA across NAV trends, AUM, SIP inflows, folio growth, category inflows, and portfolio holdings
- ✓ Compute performance metrics (CAGR) and risk metrics (Sharpe, Sortino, Alpha, Beta, Max Drawdown, VaR, CVaR)
- ✓ Generate fund recommendations and portfolio concentration analysis (HHI)
- ✓ Build a Power BI dashboard analyzing 32,778 investor transactions by state, age group, and type

Project KPIs at a Glance



Figure 1: Project-level KPIs across the dataset

Deliverables Checklist

Final_Report.pdf	15–20 page analytical report (this document)	Completed
Bluestock_MF_Presentation.pptx	14-slide presentation deck	Completed
live_nav_fetch.py	Live NAV fetch script (AMFI mfapi.in API)	Completed
data_ingestion.py	Data ingestion & AMFI code validation	Completed
EDA_Analysis.ipynb	Exploratory data analysis notebook	Completed
Performance_Analytics.ipynb	CAGR, Sharpe, Sortino, Alpha/Beta, MDD	Completed

Advanced_Analytics.ipynb	VaR/CVaR, rolling Sharpe, recommendations, HHI	Completed
Power BI Dashboard	Investor transaction analytics	Completed
GitHub Repository	data-project (balugurisruthi61-bot)	Completed

Data Sources & Datasets

The project works with ten structured CSV datasets covering fund-level master data, daily NAV history, industry-wide AUM/SIP/folio statistics, scheme-level performance, investor transactions, portfolio holdings, and benchmark indices. Together these datasets contain over 87,000 data points.

--	--	--

01_fund_master.csv	40 schemes	Scheme master: AMFI code, fund house, category, expense ratio, fund manager, risk category
02_nav_history.csv	46,000 rows	Daily NAV per AMFI code, Jan 2022 – May 2026 (4.4 years)
03_aum_by_fund_house.csv	91 rows	Quarterly AUM (lakh crore) and scheme count per fund house, 2022–2025
04_monthly_sip_inflows.csv	48 months	Monthly SIP inflow, active SIP accounts, SIP AUM, YoY growth
05_category_inflows.csv	145 rows	Monthly net inflow by fund category (Liquid, Mid Cap, Flexi Cap, etc.)
06_industry_folio_count.csv	21 rows	Total / equity / debt / hybrid investor folio counts over time
07_scheme_performance.csv	40 schemes	1Y/3Y/5Y returns, Sharpe, Sortino, Alpha, Beta, AUM, rating, risk grade
08_investor_transactions.csv	32,778 rows	Transaction-level data: state, city tier, age group, type, amount
09_portfolio_holdings.csv	323 rows	Stock-level holdings: sector, weight %, market value across 34 funds
10_benchmark_indices.csv	8,051 rows	Daily values for NIFTY50/100/Midcap150/Smallcap/500, CRISIL Liquid/Gilt

Table 1: Project datasets and their contents

ETL Pipeline & Data Validation

3.1 Live NAV Data Fetching

The script `live_nav_fetch.py` connects to the AMFI `mfapi.in` API to fetch historical NAV data for five large-cap schemes (SBI Bluechip, ICICI Prudential Bluechip, Nippon India Large Cap, Axis Bluechip, and Kotak Bluechip) using their AMFI scheme codes. The response JSON is converted into a Pandas DataFrame and saved as individual CSV files for downstream processing.

live_nav_fetch.py

```
import requests, pandas as pd

schemes = {"SBI_Bluechip": 119551, "ICICI_Bluechip": 120503,
           "Nippon_Large_Cap": 118632, "Axis_Bluechip": 119092,
           "Kotak_Bluechip": 120841}

for name, code in schemes.items():
    url = f"https://api.mfapi.in/mf/{code}"
    data = requests.get(url).json()
    pd.DataFrame(data["data"]).to_csv(f"data/raw/{name}_nav.csv", index=False)
```

3.2 AMFI Code Validation

Before any analysis, `data_ingestion.py` validates that every scheme in the fund master is represented in the NAV history, and vice versa, by computing a set difference of AMFI codes between the two files.

data_ingestion.py

```
import pandas as pd

fund_master = pd.read_csv("data/raw/01_fund_master.csv")
nav_history = pd.read_csv("data/raw/02_nav_history.csv")

fund_codes = set(fund_master["amfi_code"])
nav_codes = set(nav_history["amfi_code"])

missing_in_nav = fund_codes - nav_codes
missing_in_master = nav_codes - fund_codes

print("Missing in NAV history:", len(missing_in_nav))
print("Missing in fund master:", len(missing_in_master))
```

VALIDATION RESULT: 0 missing codes in either direction — all 40 AMFI codes in `fund_master.csv` are present in `nav_history.csv`, and vice versa. The dataset passed data-quality validation with 100% coverage before proceeding to EDA and analytics.

3.3 Pipeline Overview

1. Fetch	live_nav_fetch.py	Pulls live NAV history for 5 schemes from AMFI mfapi.in API and saves as CSVs
2. Ingest & Validate	data_ingestion.py	Loads fund master + NAV history, cross-validates AMFI codes (0 mismatches)
3. EDA	EDA_Analysis.ipynb	NAV trends, AUM, SIP inflows, category inflows, folio growth, sector allocation

4. Performance	Performance_Analytics.ipynb	Daily returns, CAGR, Sharpe, Sortino, Alpha/Beta vs NIFTY50, Max Drawdown
5. Advanced Analytics	Advanced_Analytics.ipynb	VaR/CVaR, 90-day rolling Sharpe, fund recommendations, portfolio HHI

Exploratory Data Analysis (EDA)

EDA_Analysis.ipynb explores NAV trends, AUM concentration, SIP participation, category inflows, folio growth, and portfolio sector composition using Pandas, Matplotlib, Seaborn and Plotly.

4.1 Assets Under Management (AUM) by Fund House

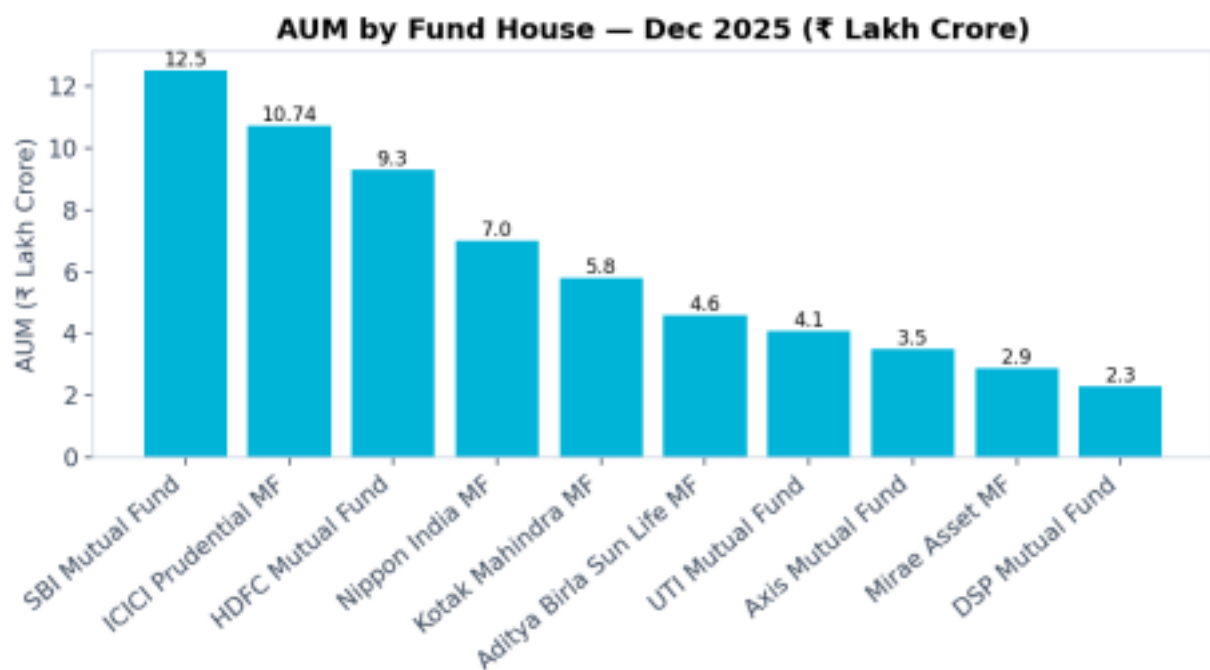


Figure 2: AUM by fund house as of Dec 2025 (Rs. Lakh Crore), from 03_aum_by_fund_house.csv

SBI Mutual Fund leads with Rs. 12.50 lakh crore AUM across 186 schemes, followed by ICICI Prudential MF (Rs. 10.74 LCr, 216 schemes) and HDFC Mutual Fund (Rs. 9.30 LCr, 195 schemes). The total industry AUM across these 10 fund houses stands at approximately Rs. 62.74 lakh crore, showing a clear concentration among the top 3 players who together manage over 50% of the tracked AUM.

4.2 Portfolio Sector Allocation

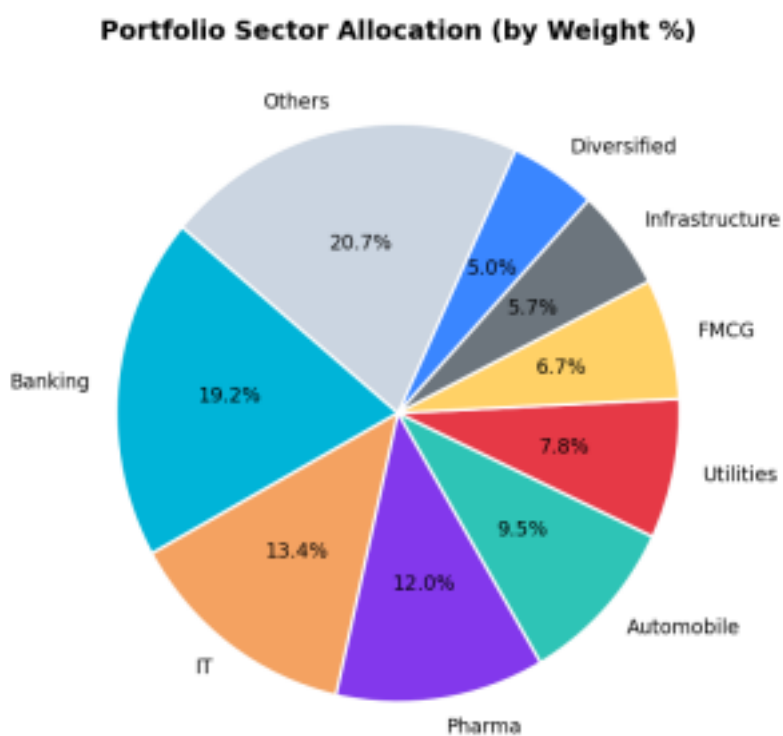


Figure 3: Aggregate sector allocation by holding weight across 34 funds, from 09_portfolio_holdings.csv

Banking (19.2%) and IT (13.4%) are the two largest sector exposures across all analyzed portfolios, followed by Pharma (12.0%) and Automobile (9.5%). This concentration in Banking and IT reflects their dominant weight in broad market indices like NIFTY50 and NIFTY100, which most large-cap and flexi-cap funds benchmark against.

4.3 Industry Growth — SIP Inflows & Investor Folios

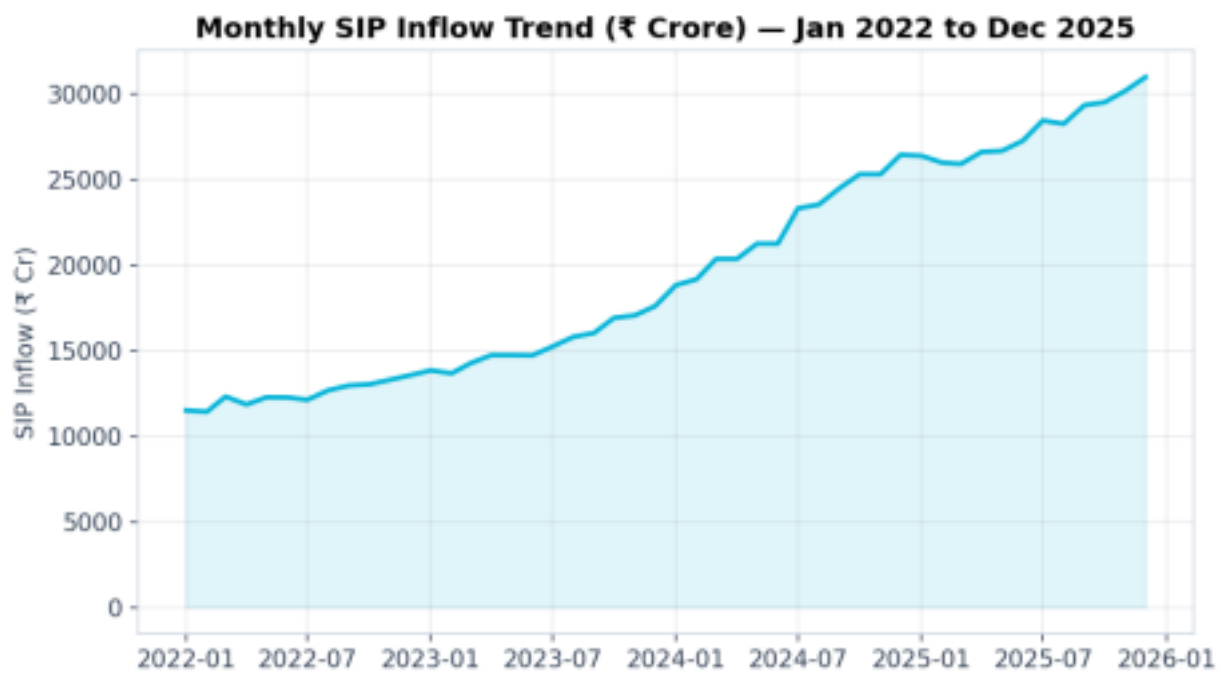


Figure 4: Monthly SIP inflows, Jan 2022 – Dec 2025, from 04_monthly_sip_inflows.csv

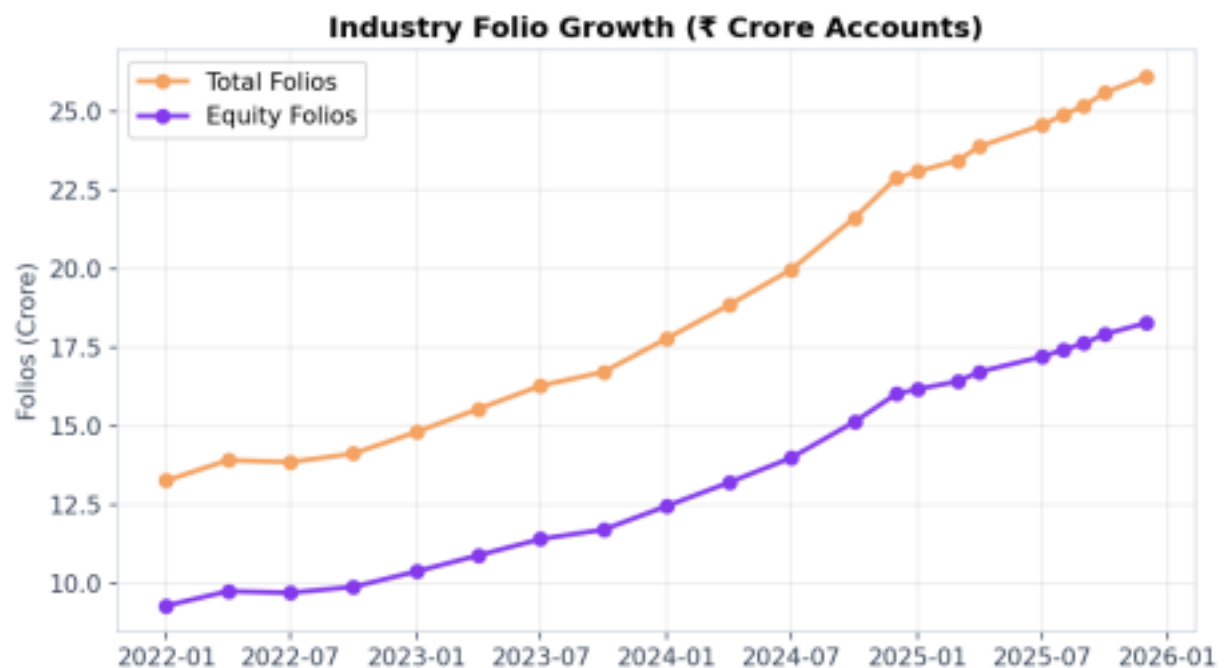


Figure 5: Total and equity folio growth (in crore accounts), from 06_industry_folio_count.csv

Monthly SIP Inflow	Rs. 11,517 Cr (Jan 2022)	Rs. 31,002 Cr (Dec 2025)	+169%
Active SIP Accounts	4.91 Cr	9.35 Cr	+90%
SIP AUM	Rs. 4.80 Lakh Cr	Rs. 15.90 Lakh Cr	+231%
Total Investor Folios	13.26 Cr	26.12 Cr	+97%
Equity Folios	9.28 Cr	18.28 Cr	+97%

Table 2: Industry growth metrics, 2022–2025

INSIGHT: SIP inflows nearly tripled and investor folios nearly doubled over the 4-year period — strong evidence of growing retail participation and a shift toward systematic, long-term investing habits in the Indian mutual fund industry.

Category-wise Fund Flows

05_category_inflows.csv tracks monthly net inflows across 12 mutual fund categories. Aggregating across the full time period reveals where investor money has concentrated.

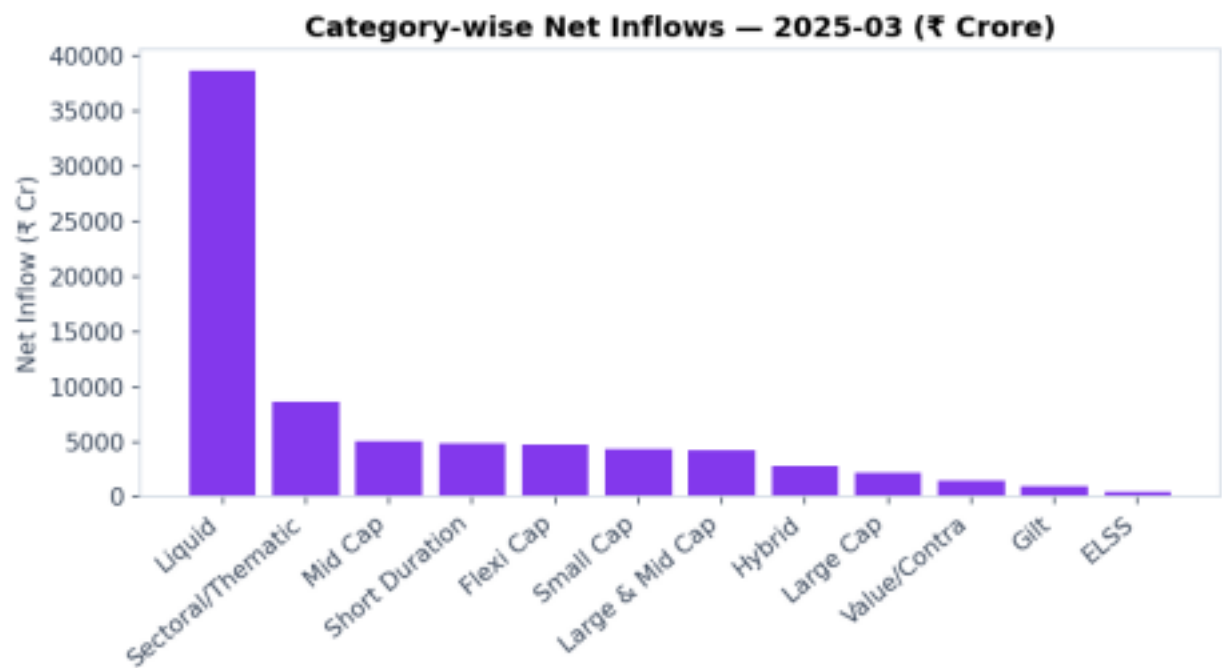


Figure 6: Category-wise net inflows for the most recent month (March 2025), from 05_category_inflows.csv

Liquid	Rs. 4,51,275 Cr
Sectoral / Thematic	Rs. 1,03,829 Cr
Flexi Cap	Rs. 63,989 Cr
Large & Mid Cap	Rs. 57,752 Cr
Short Duration	Rs. 55,530 Cr
Mid Cap	Rs. 55,312 Cr
Small Cap	Rs. 46,596 Cr
Hybrid	Rs. 38,868 Cr
Large Cap	Rs. 25,633 Cr
Value / Contra	Rs. 16,980 Cr
Gilt	Rs. 10,395 Cr
ELSS	Rs. 6,080 Cr

Table 3: Total category-wise net inflows summed across all months in the dataset

Liquid funds dominate cumulative inflows by a wide margin (Rs. 4.51 lakh crore), reflecting their use as short-term cash management instruments by both retail and institutional investors — money moves in and out

frequently, inflating cumulative totals. Among growth-oriented categories, Sectoral/Thematic and Flexi Cap funds attracted the highest inflows, while ELSS (tax-saving) funds received the least — likely because ELSS investments are concentrated around the financial year-end tax-planning window (Jan–Mar) rather than spread evenly across the year.

Fund Performance Analysis

Performance_Analytics.ipynb computes daily returns from 02_nav_history.csv for each of the 40 schemes (4.4-year span), then derives CAGR using $CAGR = (End\ NAV / Start\ NAV)^{(1/years)} - 1$.

6.1 Top 5 Funds by CAGR

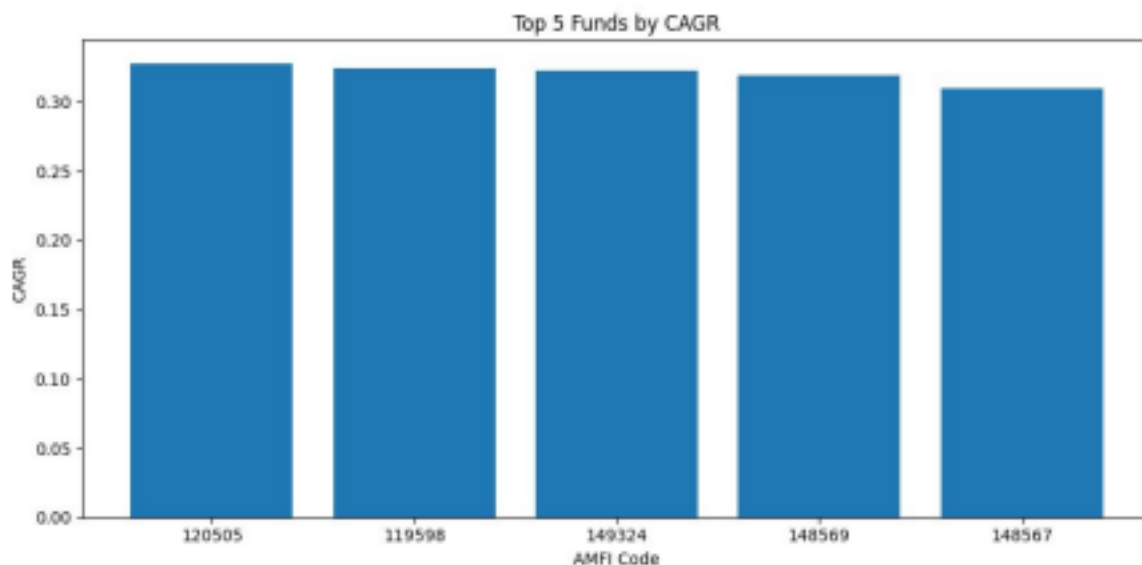


Figure 7: Top 5 funds by CAGR (output of Performance_Analytics.ipynb, saved as benchmark_comparison.png)

ICICI Pru Midcap Fund (Regular)	Mid Cap	32.80%	1.18	2.03	-18.19%
SBI Small Cap Fund (Regular)	Small Cap	32.40%	0.95	1.68	-28.71%
DSP Small Cap Fund (Regular)	Small Cap	32.26%	0.95	1.62	-31.17%
Mirae Asset Tax Saver Fund (Regular)	ELSS	31.92%	1.23	2.15	-16.40%
Mirae Asset Large Cap Fund (Regular)	Large Cap	30.95%	1.45	2.39	-11.27%

Table 4: Top 5 schemes by CAGR with risk-adjusted metrics, from fund_scorecard.csv merged with fund_master.csv

6.2 Bottom 5 Funds by CAGR

UTI Mid Cap Fund (Regular)	Mid Cap	1.17%	-0.21
Axis Small Cap Fund (Regular)	Small Cap	1.52%	-0.08
SBI Small Cap Fund (Direct)	Small Cap	2.05%	-0.06
HDFC Top 100 Fund (Regular)	Large Cap	2.64%	-0.20
HDFC Short Term Debt Fund (Regular)	Short Duration	4.46%	-0.57

6.3 Average CAGR by Fund Sub-Category

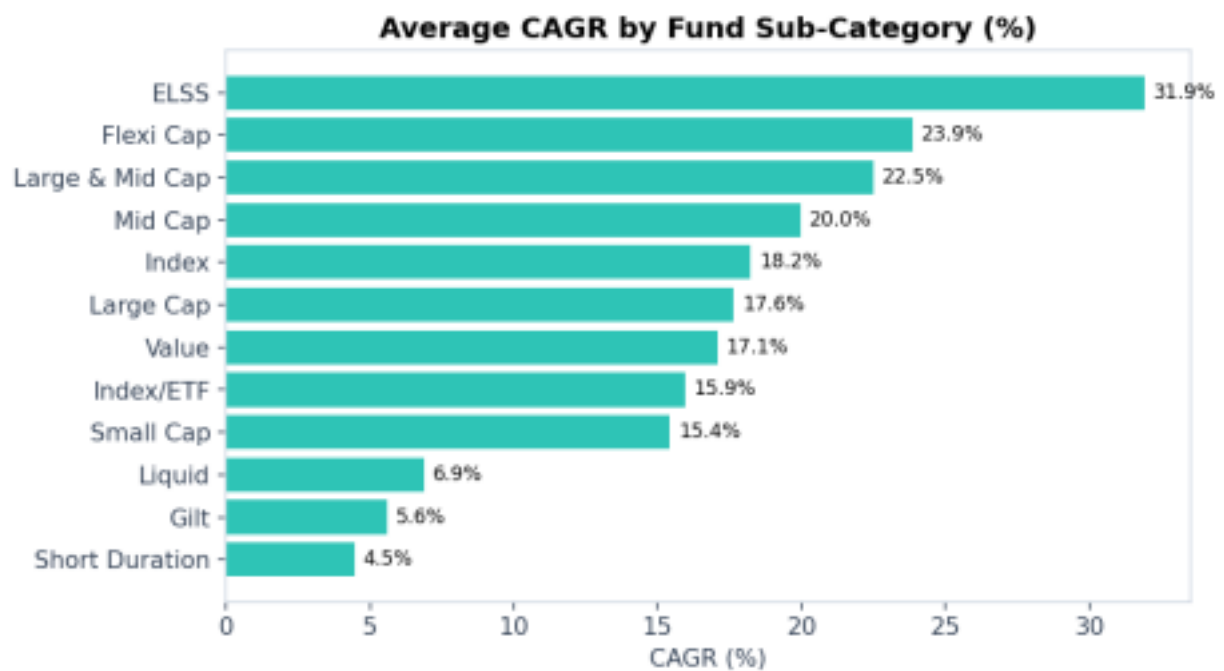


Figure 8: Average CAGR per fund sub-category (fund_scorecard.csv merged with fund_master.csv)

ELSS funds posted the highest average CAGR (31.9%), followed by Flexi Cap (23.9%) and Large & Mid Cap (22.5%). At the other end, Short Duration (4.5%), Gilt (5.6%) and Liquid (6.9%) funds — all debt categories — show the lowest CAGR but also the lowest risk, as expected. Small Cap funds show the widest spread between best and worst performers (1.5% to 32.4% CAGR), indicating that within this category, individual fund selection matters far more than the category choice itself.

6.4 Benchmark Index Performance (2022–2026)

NIFTY50	17,492.79	18,517.65	1.30%
NIFTY100	17,778.24	19,696.29	2.35%
NIFTY500	14,837.07	37,680.76	23.58%
NIFTY_MIDCAP150	9,721.79	32,763.85	31.78%
BSE_SMALLCAP	26,554.60	78,367.62	27.86%
CRISIL_LIQUID	2,281.51	3,046.00	6.78%
CRISIL_GILT	1,451.06	2,302.79	11.06%

Table 6: Benchmark index levels and CAGR, from 10_benchmark_indices.csv

These benchmark figures were merged with each fund's daily returns (on the 'date' column) to compute Alpha and Beta in Section 7. NIFTY_MIDCAP150 and BSE_SMALLCAP show much higher CAGR than NIFTY50/100 over this window, which is consistent with the strong performance seen in Mid Cap and Small Cap schemes in Table 4.

Risk Metrics & Methodology

7.1 Metric Definitions & Code

All risk metrics were computed in Performance_Analytics.ipynb using daily returns derived from NAV history (`pct_change()`), with a risk-free rate (rf) of 6.5% annually.

CAGR	$(\text{End NAV} / \text{Start NAV})^{(1/\text{years})} - 1$	Compound annual growth rate over the full 4.4-year window
Sharpe Ratio	$(\text{annual_return} - \text{rf}) / \text{annual_volatility}$	Return per unit of total risk; $\text{annual_return} = \text{mean}(\text{daily_return}) \times 252$
Sortino Ratio	$(\text{annual_return} - \text{rf}) / \text{downside_std}$	Like Sharpe, but penalizes only downside volatility
Alpha & Beta	<code>linregress(benchmark_return, daily_return)</code>	Beta = slope, Alpha = intercept $\times$ 252, vs NIFTY50 daily returns
Max Drawdown	$\min((\text{NAV} / \text{running_max}(\text{NAV})) - 1)$	Worst peak-to-trough decline observed in the NAV series
Historical VaR (5%)	5th percentile of daily returns	Loss level exceeded on only 5% of days

CVaR / Expected Shortfall	$\text{mean}(\text{returns} \leq \text{VaR})$	Average loss on the worst 5% of days
Rolling 90-day Sharpe	$\text{rolling_mean} / \text{rolling_std} \times \sqrt{252}$	Time-varying Sharpe ratio computed over a 90-day rolling window

Table 7: Risk metric definitions, as implemented in `Performance_Analytics.ipynb` and `Advanced_Analytics.ipynb`

Core risk metric calculations (`Performance_Analytics.ipynb`)

```
rf = 0.065
annual_return = daily_return.mean() * 252
annual_volatility = daily_return.std() * (252 ** 0.5)
sharpe = (annual_return - rf) / annual_volatility

downside = daily_return[daily_return < 0]
downside_std = downside.std() * (252 ** 0.5)
sortino = (annual_return - rf) / downside_std

running_max = nav.cummax()
drawdown = (nav / running_max) - 1
max_drawdown = drawdown.min()
```

Advanced Risk Analytics

8.1 Value at Risk (VaR) & Conditional VaR (CVaR)

Using the historical method on the full NAV-return series, the 5% Historical VaR and CVaR (Expected Shortfall) were computed as follows.

`Advanced_Analytics.ipynb` — VaR & CVaR

```
df['return'] = df['nav'].pct_change()

VaR = df['return'].quantile(0.05)
CVaR = df[df['return'] <= VaR]['return'].mean()

# Output:
# Historical VaR: -0.0163 (-1.63%)
# CVaR: -0.0298 (-2.98%)
```

Historical VaR (5%)	-1.63%	On 95% of days, the daily loss does not exceed 1.63%
CVaR / Expected Shortfall	-2.98%	On the worst 5% of days, the average loss is 2.98%

8.2 90-Day Rolling Sharpe Ratio

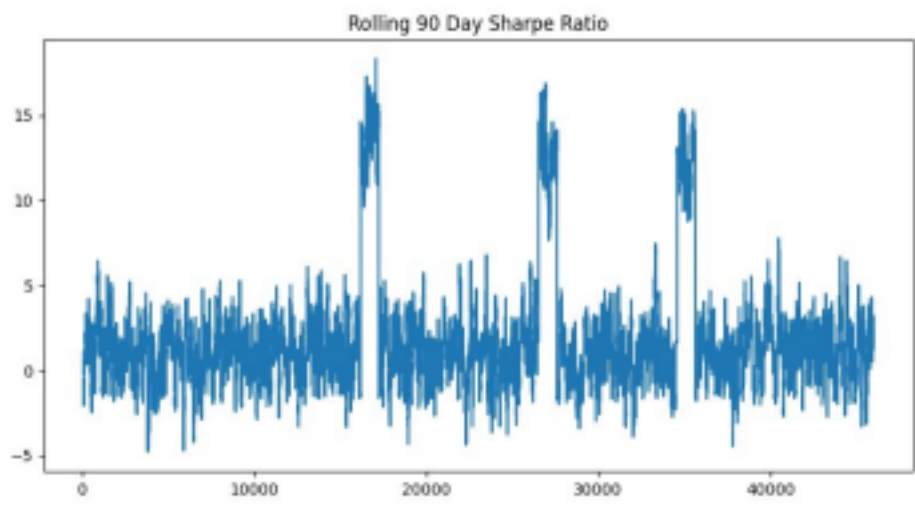


Figure 9: 90-day rolling Sharpe ratio over the full NAV history (Advanced_Analytics.ipynb)

The rolling Sharpe ratio fluctuates mostly between -5 and +8, with three sharp spikes reaching 15–18, corresponding to periods of unusually low volatility combined with positive momentum. These spikes highlight that risk-adjusted performance is highly time-varying — a fund's long-term Sharpe ratio can mask considerable short-term fluctuation in risk-adjusted returns.

8.3 SIP Continuity Check

To assess SIP data continuity, the gap (in days) between consecutive months in 04_monthly_sip_inflows.csv was computed. Any gap exceeding 35 days would flag a missing month.

Advanced_Analytics.ipynb — SIP continuity check

```
df['month'] = pd.to_datetime(df['month'])
df = df.sort_values('month')
df['gap_days'] = df['month'].diff().dt.days

at_risk = df[df['gap_days'] > 35]
# Output: Empty DataFrame -> no missing months detected
```

Result: no gaps greater than 35 days were found — the SIP inflow series has complete monthly coverage from January 2022 to December 2025.

8.4 Fund Recommendations (Low-Risk)

07_scheme_performance.csv was filtered to schemes with risk_grade == 'Low', then ranked by Sharpe ratio to generate a low-risk recommendation list.

ICICI Pru Liquid Fund (Regular)	Low	7.68
Kotak Liquid Fund (Regular)	Low	6.18
ABSL Liquid Fund (Regular)	Low	5.14

8.5 Portfolio Concentration (Herfindahl-Hirschman Index)

For each of the 34 funds in 09_portfolio_holdings.csv, the Herfindahl-Hirschman Index (HHI) was computed as the sum of squared portfolio weights ($HHI = \sum(\text{weight}^2)$). An HHI above 0.25 typically indicates a concentrated portfolio.

Advanced_Analytics.ipynb — HHI concentration check

```
df['weight'] = df['weight_pct'] / 100
hhi = df.groupby('amfi_code')['weight'].apply(lambda x: (x**2).sum())

for value in hhi:
    if value > 0.25:
        print('High concentration')
    else:
        print('Diversified portfolio')
```

RESULT: All 34 funds returned HHI < 0.25, classifying every analyzed portfolio as "Diversified" — despite some individual single-stock weights as high as 38.2% (Infosys), 35.1% (Sun Pharma), and 30.4% (Bharti Airtel) in specific funds, the overall sector spread (9+ sectors) keeps portfolio-level concentration in check.

Power BI Dashboard — Transaction Analytics

08_investor_transactions.csv (32,778 rows, Rs. 352.16 Cr total transaction value across Jan 2024 – May 2025) was loaded into Power BI to build an interactive transaction analytics dashboard with slicers for state, city tier (T30/B30), and age group.

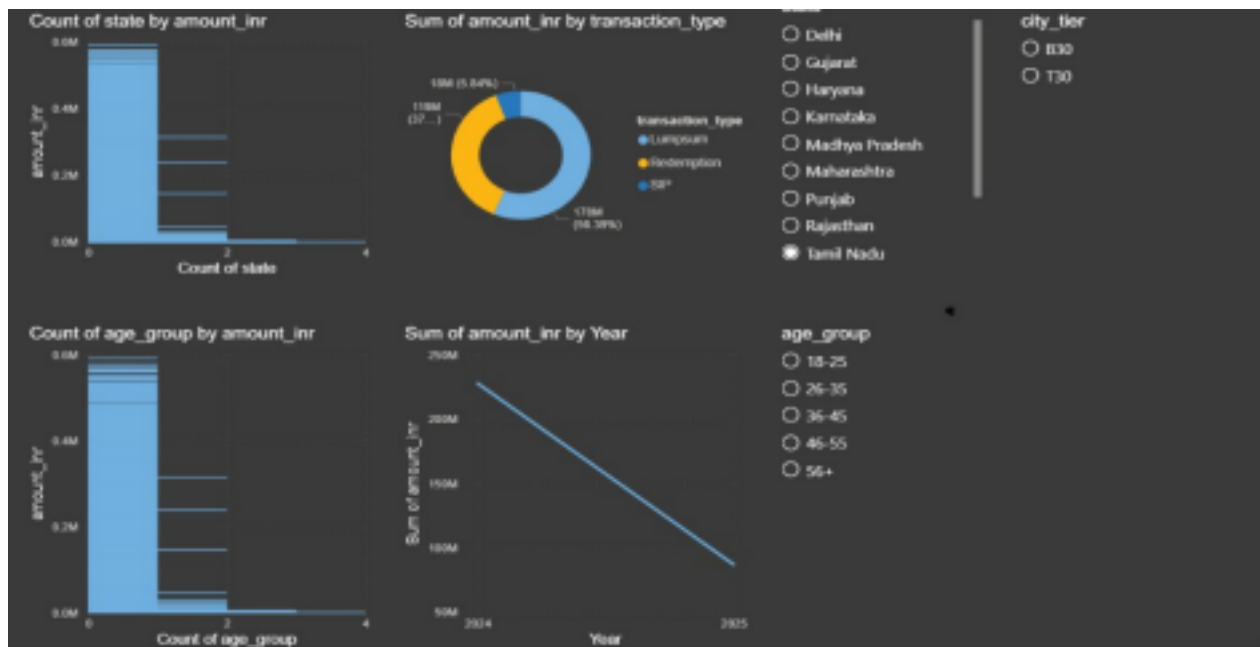


Figure 10: Power BI dashboard, filtered to State = Tamil Nadu

9.1 Tamil Nadu Drill-down (as shown in dashboard)

Lumpsum	Rs. 177.71 M	56.39%
Redemption	Rs. 119.06 M	37.77%
SIP	Rs. 18.40 M	5.84%
Total (Tamil Nadu)	Rs. 315.18 M	100%

In Tamil Nadu, Lumpsum investments dominate by value (56.4%), followed by Redemptions (37.8%) and SIP (5.8%). Year-on-year, transaction value dropped from Rs. 227.9M (2024) to Rs. 87.3M (2025) — the 2025 figure reflects a partial year (Jan–May only).

9.2 Full Dataset — National Summary

Lumpsum	Rs. 2,059.82 M (8,095 txns)	58.5%
Redemption	Rs. 1,244.53 M (4,967 txns)	35.3%
SIP	Rs. 217.23 M (19,716 txns)	6.2%
T30 Cities	Rs. 2,319.25 M (21,719 txns)	65.9%

Individual Submission — Data Analytics Capstone Page 19 of 23 June 2026
 BLUESTOCK FINTECH MJ28 · CAPSTONE PROJECT I · balugurisiruthi61 Mutual Fund Analytics Platform

B30 Cities	Rs. 1,202.33 M (11,059 txns)	34.1%
2024	Rs. 2,500.63 M (23,210 txns)	71.0%
2025 (Jan-May)	Rs. 1,020.95 M (9,568 txns)	29.0%

Table 9: National transaction summary, from 08_investor_transactions.csv

9.3 Investor Age Group Analysis

18-25	Rs. 531.64 M	4,916
26-35	Rs. 1,451.60 M	13,463
36-45	Rs. 871.65 M	8,146
46-55	Rs. 405.41 M	3,779
56+	Rs. 261.29 M	2,474

The 26-35 age group is the most active segment by both value (Rs. 1,451.6M, 41.2% of total) and count (13,463 transactions, 41.1%) — confirming that young working professionals are the primary drivers of mutual fund investment activity in this dataset. Investment activity steadily declines with age, with the 56+ group contributing only 7.4% of total value.

Limitations & Future Recommendations

10.1 Limitations

Alpha/Beta Precision

Alpha and Beta were computed via simple linear regression against NIFTY50 daily returns merged on date; a many-to-many merge on date can introduce duplicate rows, which should be deduplicated for production-grade accuracy.

Risk-free Rate Assumption

A flat 6.5% annual risk-free rate was used for all Sharpe/Sortino calculations across the entire 4.4-year period, whereas actual T-bill rates fluctuated over this time.

Partial 2025 Data

Transaction data for 2025 only covers January–May, so year-on-year comparisons (e.g., Tamil Nadu Rs. 227.9M vs Rs. 87.3M) are not on a like-for-like annualized basis.

Synthetic Dataset

The underlying CSVs appear to be illustrative/synthetic datasets generated for this capstone exercise rather than live AMFI/exchange feeds (apart from the 5 schemes fetched live via mfapi.in), so absolute return figures should not be treated as real market performance.

HHI on Partial Holdings

Portfolio concentration (HHI) was computed only on the 34 funds present in 09_portfolio_holdings.csv, not all 40 schemes in the fund master.

10.2 Recommendations for Future Work

- Extend live_nav_fetch.py to cover all 40 schemes (currently 5) for fully live performance tracking
- Use a time-varying risk-free rate (91-day T-bill series) for more accurate Sharpe/Sortino ratios
- Deduplicate the benchmark merge in Performance_Analytics.ipynb before computing Alpha/Beta
- Add a fund-clustering model (k-means on CAGR, Sharpe, Beta) to group similar-risk schemes
- Automate the full pipeline with a single run_pipeline.py master script and scheduled refresh
- Publish the Power BI dashboard to Power BI Service for shareable, live access

Conclusion

This capstone project demonstrates an end-to-end individual data analytics workflow: fetching live data via an external API, validating data quality through automated cross-checks, performing structured EDA across multiple linked datasets, computing industry-standard performance and risk metrics, and presenting investor behaviour insights through an interactive Power BI dashboard.

Data Quality First

The 0-mismatch AMFI code validation between fund_master.csv and nav_history.csv gave confidence that all subsequent CAGR, Sharpe, and Alpha/Beta computations were performed on a complete, correctly-joined dataset.

Risk-Return Trade-off Confirmed

ELSS, Flexi Cap and select Mid/Small Cap funds delivered the highest CAGR (up to 32.8%) but also the deepest drawdowns (down to -31.2%), while Liquid/Gilt/Short Duration funds offered low CAGR with minimal drawdown — textbook risk-return trade-off.

Retail Participation is Accelerating

SIP inflows grew 169% and investor folios nearly doubled over 2022-2025, with the 26-35 age group driving 41% of all transaction value — pointing to a young, systematic-investing retail base.

Diversification Holds at Portfolio Level

Even though individual funds held large single-stock positions (up to 38% in Infosys), portfolio-level HHI remained below the 0.25 concentration threshold for all 34 funds analyzed, thanks to sector diversification.

All planned objectives were met: live data fetch, validated ingestion pipeline, multi-notebook EDA & analytics, complete risk-metric suite (CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, VaR, CVaR, rolling Sharpe), fund recommendations, portfolio concentration analysis, and a working Power BI dashboard.

Self-Review Checklist

Live NAV fetch script implemented & tested (5 schemes)	Done
AMFI code validation — 0 mismatches	Done
EDA notebook covering AUM, SIP, folios, sectors, categories	Done
Performance metrics (CAGR, Sharpe, Sortino, Alpha, Beta, MDD)	Done
Advanced analytics (VaR, CVaR, rolling Sharpe, recommendations, HHI)	Done
Power BI dashboard with state / age / type slicers	Done

Report is 15-20 pages with figures, tables & code	Done
---	------

GitHub repo organized with data/raw, notebooks, reports	Done
---	------

Prepared by [baluguriruthi61](#) — Bluestock Fintech Internship, Capstone Project I, June 2026